

Ruidong Zhang

M.S. Student in Computer Science

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Summary

Master's student at the University of Science and Technology of China working on compiler systems, GPU and AI systems, and hardware-software co-design.

Education

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| M.S. | University of Science and Technology of China (USTC) , Computer Science | Hefei, China
Sept 2025 – present |
| B.S. | University of Electronic Science and Technology of China (UESTC) , Computer Science
• GPA: 3.84 / 4.00 | Chengdu, China
Sept 2021 – June 2025 |

Experience

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| Ubiquant Technology , Quantitative Implementation Intern — LLM Systems | Shanghai, China
Aug 2026 – Sept 2026 |
| <ul style="list-style-type: none">• Worked on multi-node, multi-GPU LLM inference systems, including prefill-decode disaggregation and distributed execution.• Used NVIDIA profiling tools to analyze GPU kernels and system-level bottlenecks, focusing on compute, memory, and communication behavior. | |
| Shanghai Artificial Intelligence Laboratory , Research Intern — High-Performance Compilation | Shanghai, China
Jan 2026 – July 2026 |
| <ul style="list-style-type: none">• Worked on MLIR-based compiler infrastructure and performance optimization for GPU-intensive workloads, covering compiler IRs, transformations, lowering, and GPU code generation.• Implemented and evaluated compiler transformations related to program rewriting and memory-system optimization.• Profiled generated CUDA workloads across GPU platforms using low-level performance analysis, studying memory traffic, synchronization, parallelism, and occupancy bottlenecks. | |
| Houmo Technology , AI Compiler Development Intern | Nanjing, China
Apr 2025 – July 2025 |
| <ul style="list-style-type: none">• Implemented 44 ONNX operators end-to-end, from MLIR dialect definitions and pattern rewriting to hardware-specific backend primitives.• Built an automated compiler-validation pipeline with generated operator tests and ONNX Runtime golden outputs for end-to-end correctness checking. | |

Projects

RISC-V Processor Design & Validation

Designed and implemented a five-stage RV32E/RV32I processor and deployed it on an Artix-7 FPGA to boot RT-Thread.

- Added forwarding, hazard handling, CSR/exception/interrupt support, and an AXI4-based memory subsystem.
- Developed an RV32IM emulator and instruction-level differential-testing infrastructure against Spike.

Technical Skills

Programming: C++, C, Python, CUDA C++, Verilog

Compilers & Systems: MLIR, LLVM, CUDA, PyTorch, ONNX, Linux

Profiling & Development: NVIDIA Nsight Compute, Git, Docker, Verilator, Vivado

Selected Honors

- IEEEExtreme Programming Competition — Top 5% Worldwide
- HP Responsible Entrepreneurship Competition — First Prize